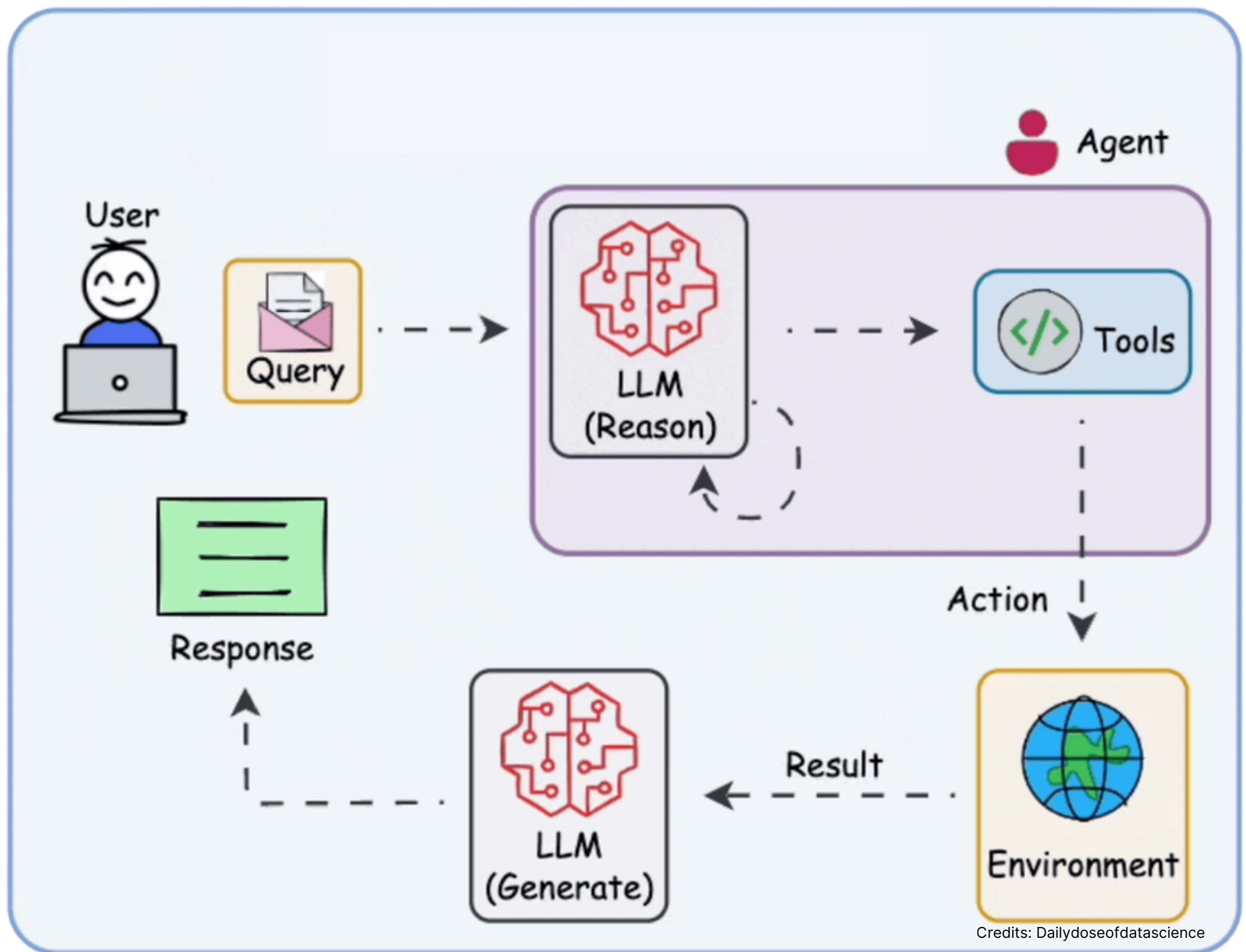


The Agent Loop



#30DaysofAgenticAI

What is the Agent Loop?

At the heart of agentic AI is a simple but powerful cycle:

Think → Act → Observe.

This loop is inspired by the **ReAct framework**, which blends reasoning and acting into iterative steps. Instead of producing a single output and stopping, an agent continuously cycles through thought, action, and feedback until the goal is reached.

How It Works

Think: The agent generates an internal “thought” — reasoning about the current state, planning next steps, or breaking down the problem. This isn’t shown to the user but guides the decision process.

Act: Based on its reasoning, the agent performs an action. This could mean calling an external tool, querying a database, running code, or generating content.

Observe: The agent then examines the result of its action. Did the tool return useful data? Did the plan succeed? This observation becomes input for the next cycle.



Why It Matters

- **Feedback Loops for Self-Correction:** Agents can detect when a step failed and adjust strategy instead of breaking.
- **Adaptability:** Each cycle incorporates new information, making the system resilient in dynamic environments.
- **Transparency:** The loop makes reasoning more traceable, giving humans insight into how decisions are made.